



Internet of Things (IoT)

DAY 5

Build your own device, then see where IoT lives

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KAUST Academy, KAUST

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On Today's Agenda

1. 🎓 **The brief:** one sensor, one actuator, one service, starting from the template
2. 🔧 **Build:** your own device, in pairs
3. 🎤 **Demos:** three minutes per pair, in front of the room
4. **IoT is everywhere:** the sectors, from the home to the factory to the city
5. **Edge computing:** when the cloud is too far away
6. **Where to go next:** single-board computers, choosing the right brain, and self-hosting your own broker and dashboard
7. **Benefits & concerns:** the promise and the price
8. **Send-off:** real stories of privacy, security, and equity in IoT, and questions worth sitting with

A simple thing that **fully works** beats an ambitious thing that does not.

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Capstone

Design and demo an IoT device of your own choosing, starter sketch and rubric included.

- Combine a sensor, an actuator, and a network service into one device
- Scope a build to the time available
- Start from a template and extend it, the way most embedded work begins
- Debug a multi-part system by testing one piece at a time

Key pins: your choice



Open the guide ↗

The Brief

In teams, combine the week into one small end-to-end device that does all three:

Sense

DHT11, PIR, LDR, pot

Act

LED, RGB, buzzer, relay, OLED

Connect

Ubidots and/or Telegram

Idea	Sense	Act	Cloud
Smart room monitor	DHT11 + LDR	OLED readings	Ubidots charts
Motion security node	PIR	Buzzer	Telegram alert
Comfort meter	DHT11	RGB green/yellow/red	Ubidots dashboard

One session to build, then a short demo. A simple thing that fully works beats an ambitious thing that does not.

Start From the Template: Three TODOs

The starter is Project 11 hollowed out. It compiles and publishes `0.0` as-is, so the **cloud half works before you wire a sensor**.

```
float readMySensor() {           // TODO 1: your sensor (ADC pin, DHT, ...)
    return 0.0;
}

ubidots.add(VAR_SENSOR, value);   // TODO 2: publish each value you care about
ubidots.publish(DEVICE_LABEL);

void callback(char *topic, byte *payload, unsigned int length) {
    digitalWrite(OUTPUT_PIN, v >= 0.5); // TODO 3: decide what a command means
}
```

- The template already does Wi-Fi, the broker, reconnect, the non-blocking publish timer, and `ubidots.loop()`.
- A new sensor is **three edits**: the `#include`, the object, and the `begin()` in `setup()`. Forgetting the third is the usual bug.
- A command payload is a **number**, so it can be a brightness, a target, or an index, not just on/off.

Scope It, Then Demo It

Finish on time

- Wire and test **one part at a time**
- Confirm good values on Serial before adding the cloud
- `millis()` , never `delay()` , so Wi-Fi stays responsive
- Keep a known-working version before the next feature

The demo (about 3 min)

- What problem does it solve, and for whom?
- Show it live: trigger the sensor, watch it respond
- One thing that was harder than expected
- One thing you would add with another day

The Week in One Loop

Every project was one idea. The capstone is where they stop being separate:

- **Read the world, decide, act** (P1): everything since is a richer version of that loop
- **Digital or analog, counts or units** (P2, P3, P9): know which sensor you have
- `millis()` , **never** `delay()` (P4): what lets a device do two things at once
- **The board can serve a page** (P5 to 9): great locally, invisible from outside
- **Keep state honest** (P8), and **local output has its place** (P10)
- **Publish/subscribe reaches anywhere** (P11), and **an existing app can be the UI** (P12)

The habits matter as much as the APIs: check hardware started, guard bad readings, keep credentials private, and print enough on Serial to see what is happening. The next device is the same shape, with different parts.

Live preview, view online at the workshop site



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GPIO Pinout

The full ESP32 DEVKIT V1 pin table: safe pins, strapping pins, input-only pins, plus ADC, touch, DAC, and the default bus pins.

- Which GPIOs are safe to use freely
- Which pins must be left alone at boot (strapping pins)
- Which pins are input-only, and which carry ADC/DAC/touch
- Default I²C and SPI bus pins



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Components Glossary

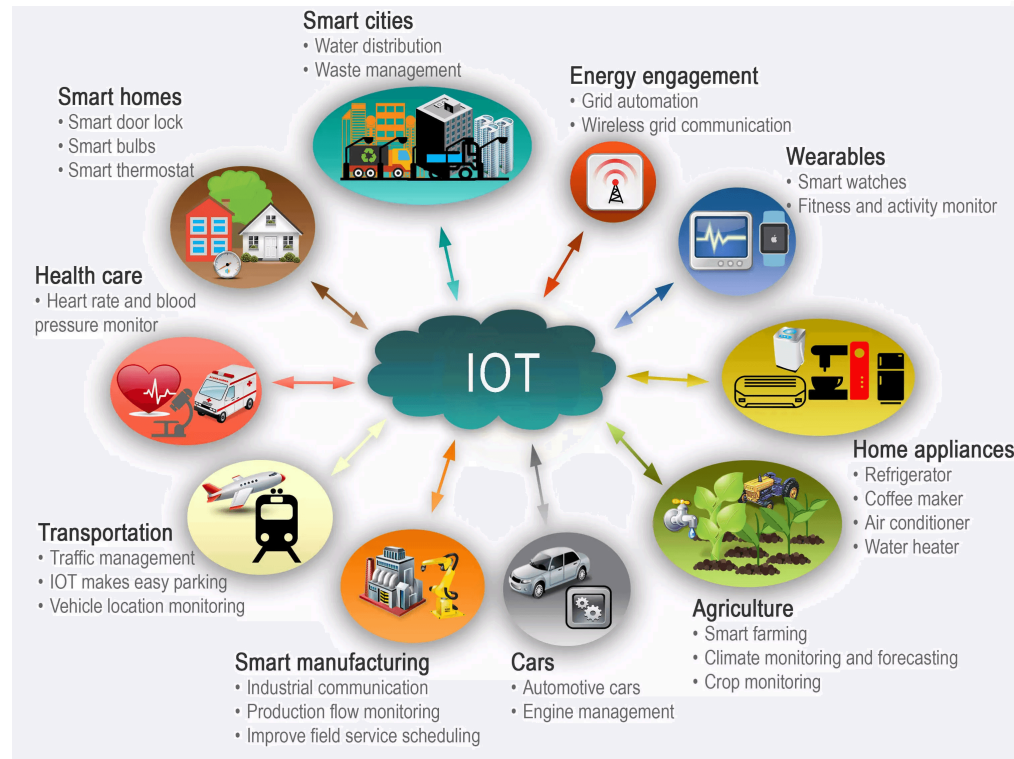
Every part in the kit: how to identify it, whether it's polarized, and which project uses it.

- A photo and description of each component in the kit
- Polarity called out where it matters
- A pointer to the project that first introduces each part



Open the guide ↗

IoT Is Everywhere



Consumer IoT

IoT devices you buy and use around the home

Consumer IoT



Devices consumers buy for everyday life: some **incredibly useful**, some **questionable**.



Around the Home



Smart speaker



Thermostat



Robot vacuum



Smart plug



On the Person



Fitness tracker



Smart watch

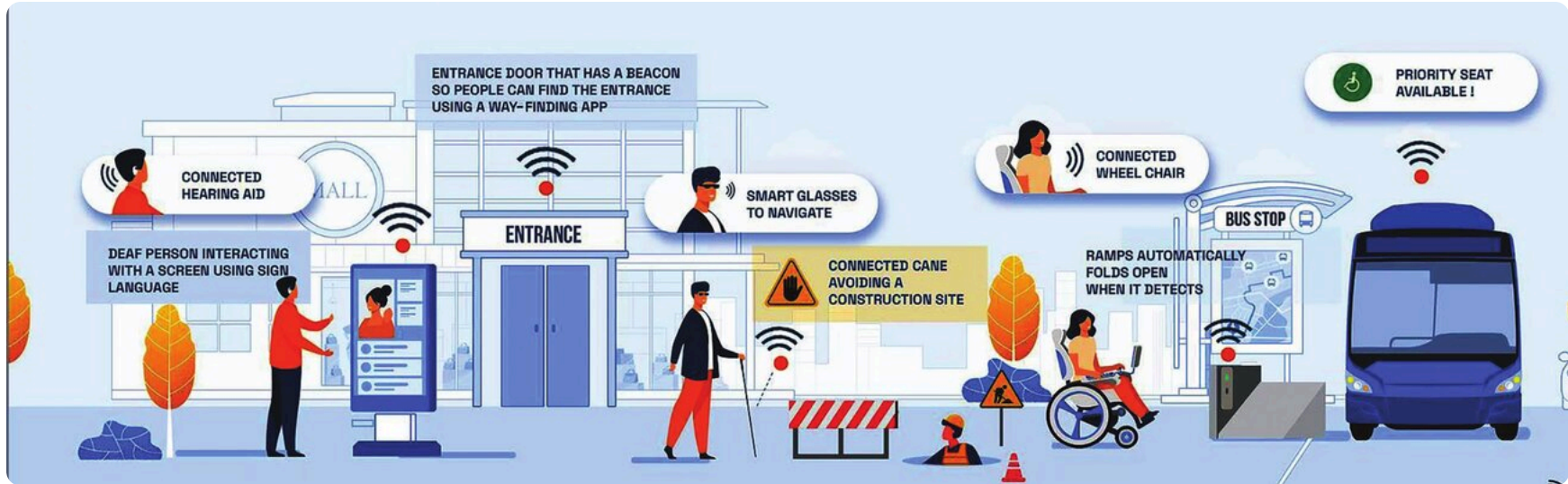


Doorbell / camera

📌 Important

Consumer IoT empowers the **1 billion people with a disability**: voice-controlled ovens, robot vacuums, and self-monitored health conditions.

Accessible IoT, in the Wild



Connected hearing aids, smart glasses, canes, and wheelchairs: the same consumer IoT building blocks, aimed at accessibility.

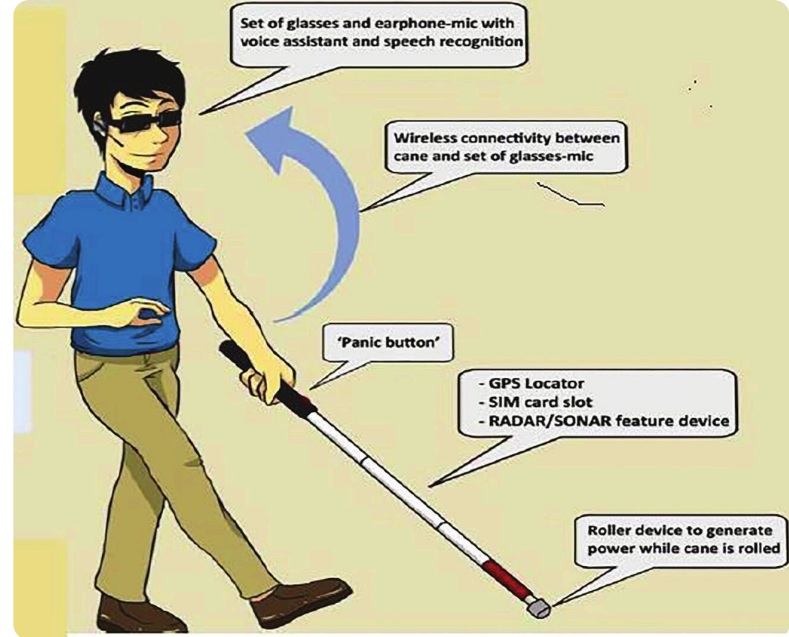
Wearables Built for Accessibility

- A Braille display on a watch face, refreshed by tiny raised pins
- Reads out notifications, time, and messages by touch, no screen needed
- Same wrist-worn form factor as any smartwatch, same sensors and connectivity underneath



The Smart Cane

- Glasses and an earphone-mic add a voice assistant and speech recognition
- The cane itself carries a GPS locator, a SIM card slot, and a RADAR/SONAR obstacle sensor
- A panic button for emergencies, and a roller in the tip that generates power as the cane rolls
- Cane and glasses talk to each other wirelessly, sharing what each one senses



The Finger Reader

- A ring-sized wearable with a camera, worn on the index finger
- Scans printed text as the finger runs along a line, word by word
- Converts what it sees to speech in real time, no phone or scanner needed
- Built to make ordinary printed material, books, mail, labels, accessible on the spot



Commercial IoT



IoT in the **workplace**, retail, and transport

Commercial IoT

 Using IoT across offices, factories, retail, and logistics.



Buildings & Retail



Occupancy
sensors



Factory
safety



Cold
storage



Smart
shelves



Transport & Logistics



Location
tracking



Mileage
charging



Hours
compliance



Arrival
alerts

Note

The goal: **cut cost and carbon**, improve **safety**, and keep operations running smoothly.

Transport & Logistics in Practice



Fleets and traffic infrastructure share live location and status data.



Tagged pallets and forklifts give warehouses real-time stock visibility.

Infrastructure IoT



Smarter **cities**, power, and environment

Infrastructure IoT

 Monitoring and managing the infrastructure people use every day.

Smart Cities



Pollution
monitoring



Traffic & parking



Public transport

Utilities



Smart power
grids



Usage analytics

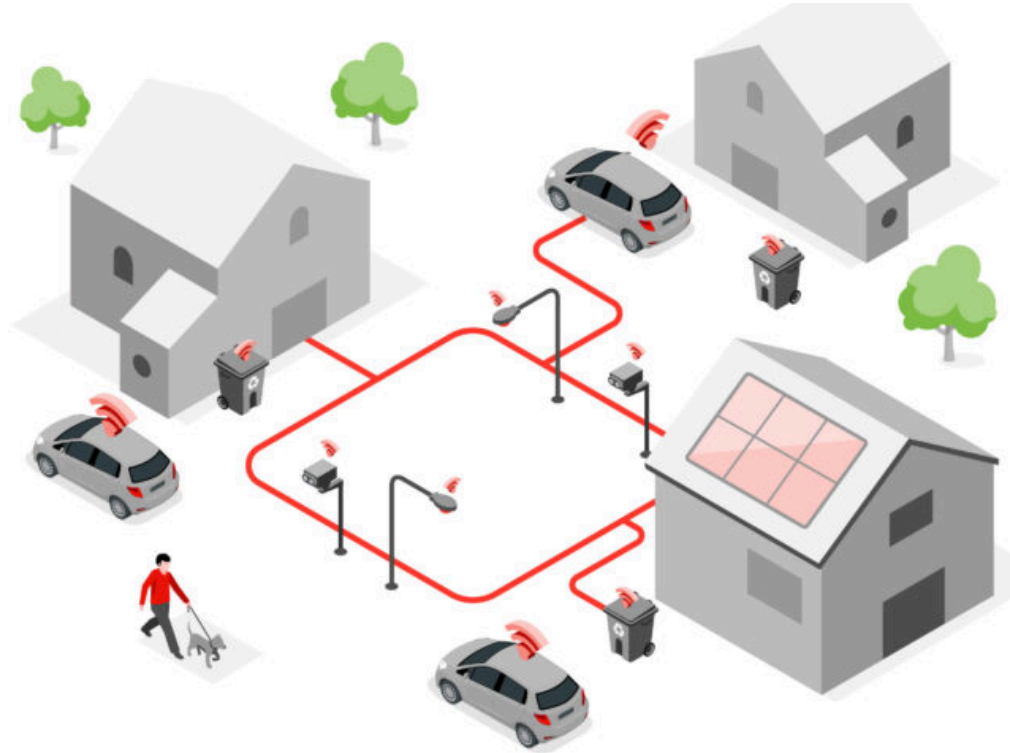


Water
monitoring

Real example: Copenhagen

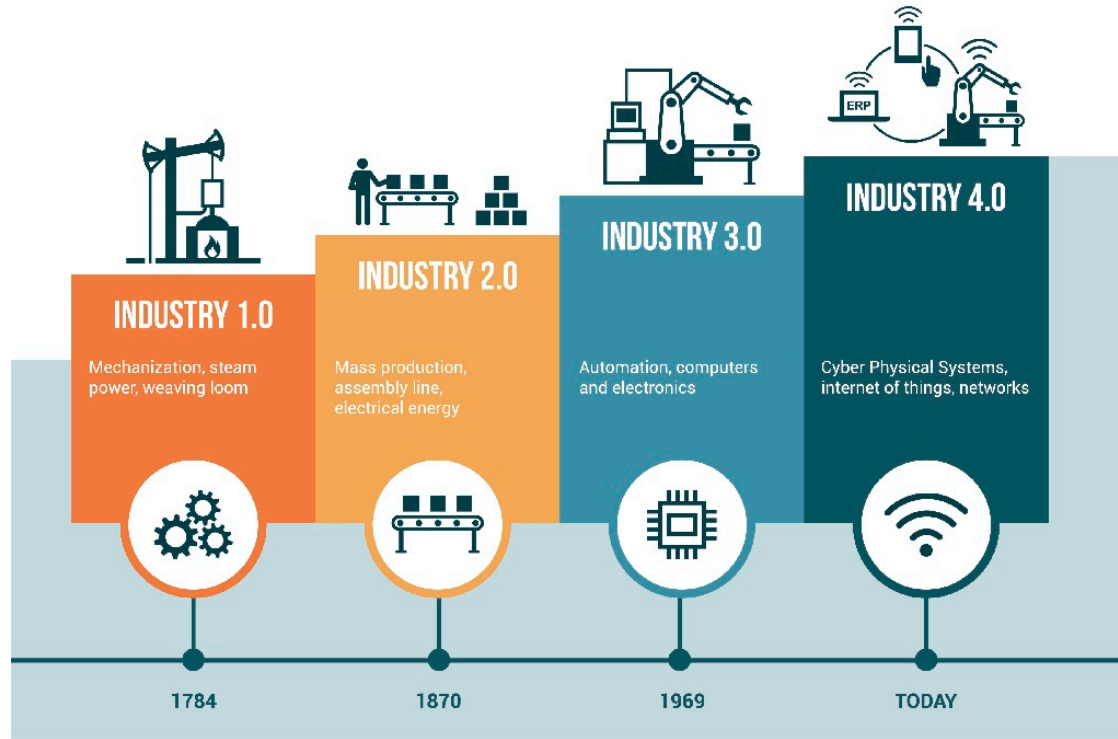
Air pollution is measured across the city and used to suggest the **cleanest cycling and jogging routes**.

A City Full of Sensors



Streetlights, bins, and cars all report status over the same connected network.

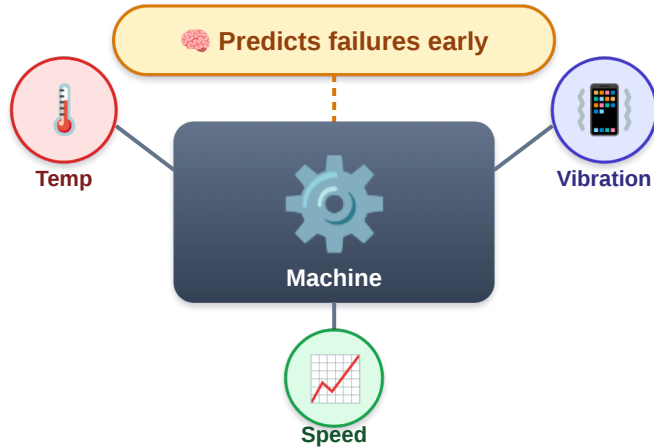
Industry 4.0 and Industrial IoT (IIoT)



From Factories to Farms



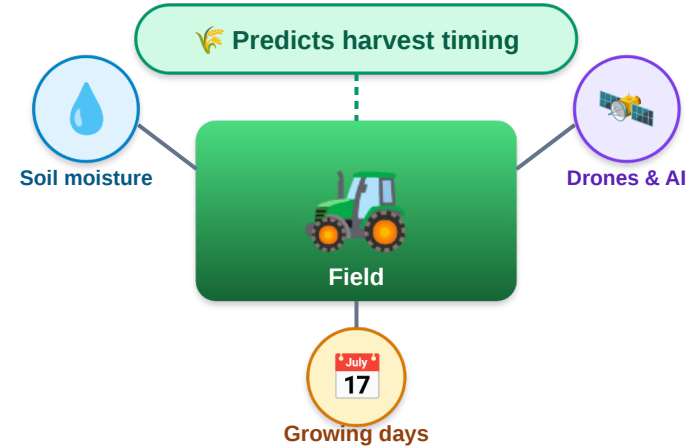
Factories



- Auto-stop when outside tolerances
- **Predictive maintenance**: AI predicts failures *before* they happen

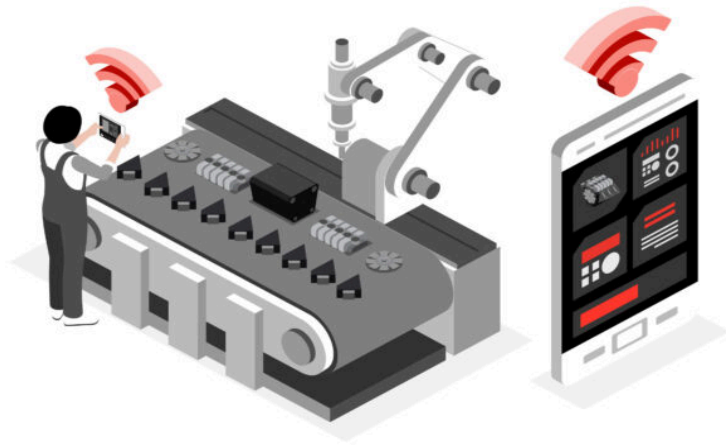


Digital agriculture

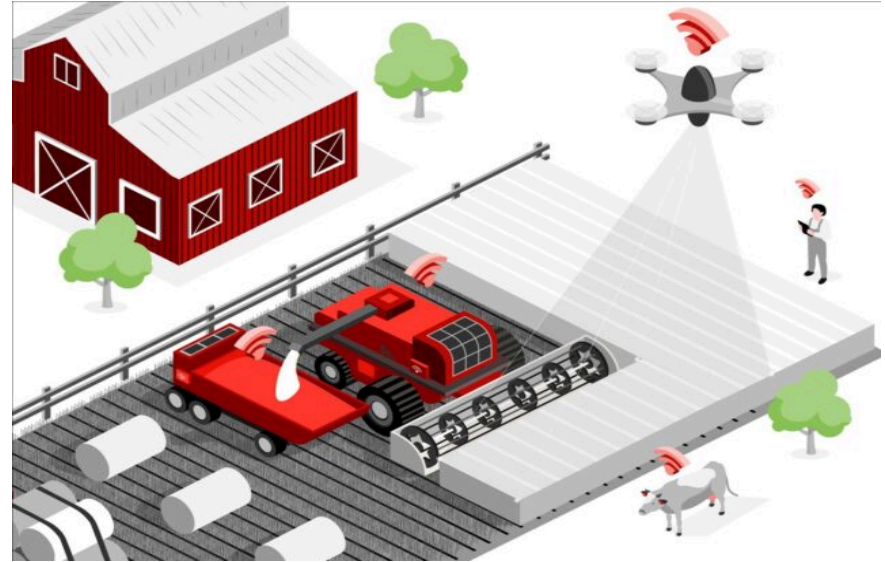


- Soil-moisture → automated watering
- **Growing degree days** to predict harvest

Industry 4.0 in Practice

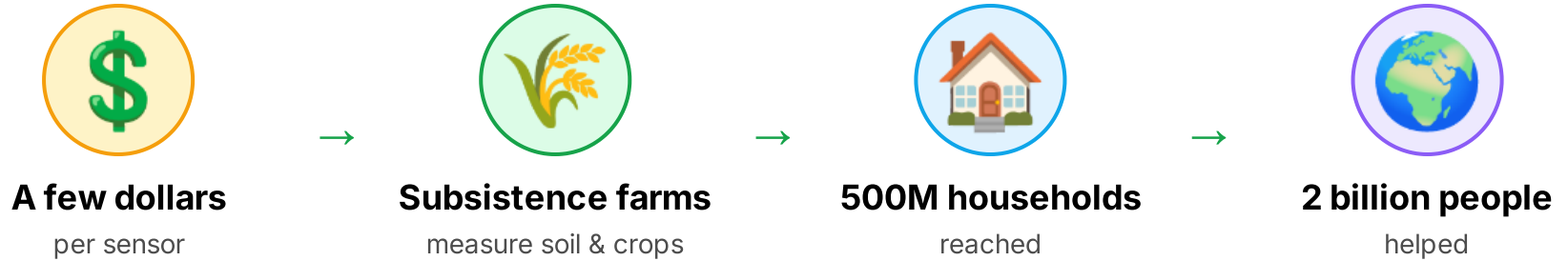


Machine data streams to a dashboard the operator carries on the floor.



Drones and sensors track harvest machinery and livestock in the field.

Why IIoT Matters



📌 Important

Digital agriculture can help the **2 billion people** in 500 million households living on subsistence farming, starting with sensors that cost just a few dollars.

Think

What other IoT devices could help farmers grow more with less?

Edge Computing

Process data **locally** instead of always calling the cloud

What is the Edge?

Even though the **I** in IoT is "Internet," devices don't always have to use it. **Edge devices** are gateways on your **local network** that process data without a round-trip to the cloud.

⚡ Speed

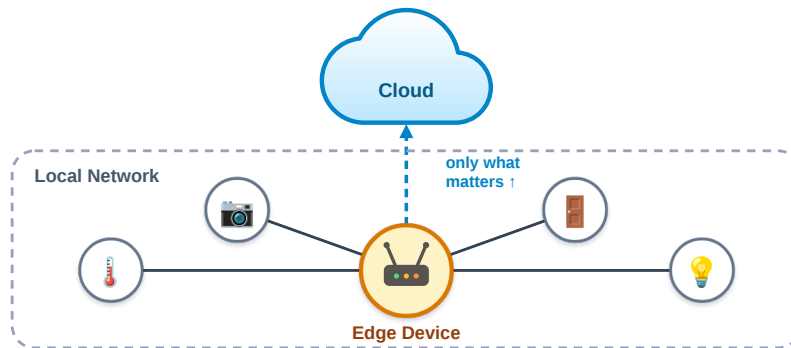
Handle lots of data or a slow connection without waiting on the Internet.

📴 Offline

Keep working with no connectivity: on a ship, or in a disaster area.

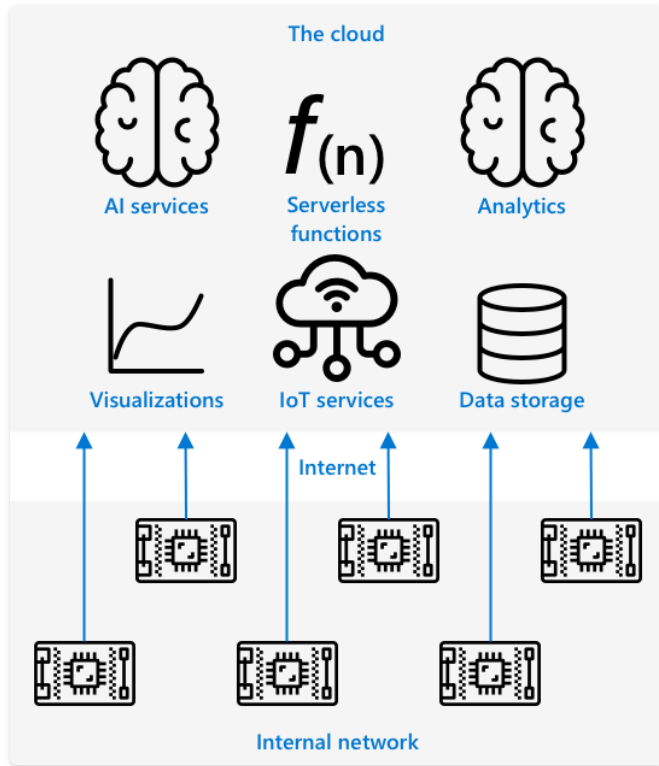
🔒 Privacy

Sensitive data stays on the local device.

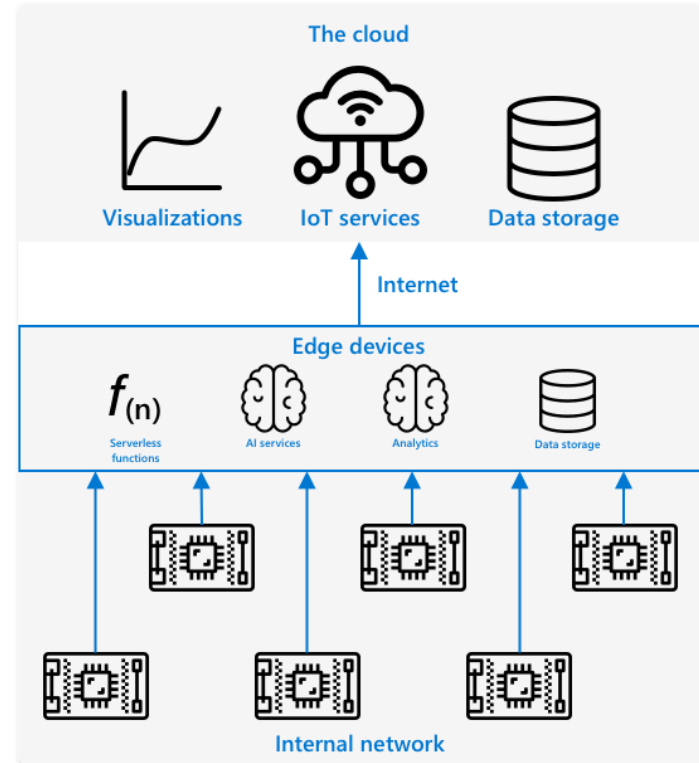


Devices talk to the edge device on the local network; only what matters goes up to the cloud.

Without the Edge vs With the Edge

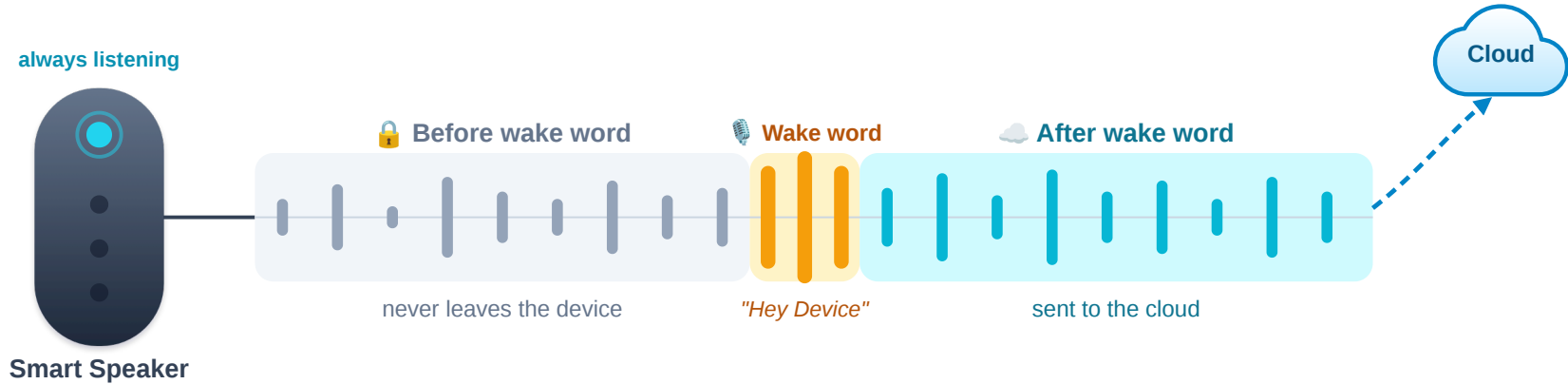


Without: every device crosses the Internet for every decision.



With: an edge device runs the same services **on your network**.

Edge in Action



A smart speaker listens for its **wake word** locally, only sending your speech to the cloud *after* it wakes.

Important

Everything said **before** the wake word (and **after** it stops listening) never leaves the device, keeping it private.

Single-Board Computers (SBCs)



A **full computer** on a single board

MCU or SBC?



Single-Board Computer

A complete computer on one board, running a **full operating system**.

- Runs **Linux**, coded in **Python**
- More power, memory & complexity
- Connect monitors, keyboards, USB
- e.g. **Raspberry Pi**



Microcontroller

A chip built to do **one specific task**.

- Runs **C/C++**, no full OS
- Tiny power draw, low cost
- Real-time, single-purpose
- e.g. **ESP32**

Side by Side



Microcontroller



Single-Board Computer

Cost	Cents to a few dollars	Tens of dollars
Power	Months on a battery	Usually mains-powered
Software	C/C++, no full OS	Linux + Python, full OS
Complexity	Single dedicated task	Many apps at once
Timing	Deterministic, real-time	OS-scheduled

Which Do I Pick?

① Reach for an MCU

Battery-powered, dedicated, real-time tasks.



Sensor node



Nightlight



Wearable

① Reach for an SBC

Complex logic, multiple peripherals, a rich UI, or on-device AI.



Rich UI



On-device AI



Many peripherals

💡 Tip

Many real products use **both**: an SBC coordinating several MCUs.



SBC
Coordinator



MCU: sensor node



MCU: motor control



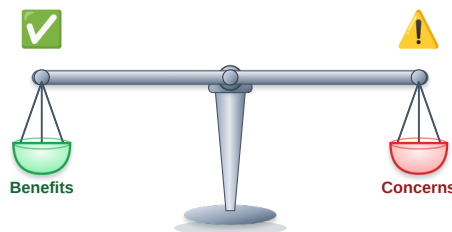
MCU: display driver

Benefits & Concerns

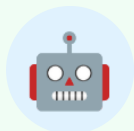


The **promise** and the **price** of connecting everything

Weighing IoT



✓ Benefits



Convenience
& automation



Cost &
energy
savings



Safety &
accessibility



Data-driven
insights

⚠ Concerns



Privacy:
who's
watching?



Security
vulnerabilities



Depends on
connectivity



Interoperability
between
brands

Social & Ethical Issues in IoT

The impact doesn't stop at the network layer

Four Recurring Tensions

Privacy & surveillance

what a device records, and who gets to hear or see it

Security breaches

the weakest device in a network can compromise the whole system

Equity & access

who benefits from smart devices, and who is left out or targeted

Obsolescence & consent

what happens when support ends, and who agreed to the data collection

Privacy & Surveillance

- **Ring doorbells and police:** Amazon's Ring partnered with police departments, including guidance on what officers should say when requesting footage from users (*The Guardian*, 2019)
- **Voice assistants, human listeners:** contractors reviewing Siri recordings "regularly heard confidential details" never intended for a person to hear (*The Guardian*, 2019)

Note

Neither example required a bug or a hack. Both worked exactly as designed. The ethical question is whether "as designed" was the right design.

Equity, Access, and Obsolescence

- **Digital redlining:** AT&T was accused of denying high-speed internet upgrades to low-income neighborhoods in Cleveland, leaving smart-home and IoT services effectively out of reach there (*cleveland.com*, 2017)
- **Health-wearable equity:** fairness and bias concerns in wearable health devices, whose training data and design choices don't always represent everyone equally (*Journal of Information Policy*, 2018)
- **Bionic eyes, discontinued:** a company stopped supporting its implanted bionic-eye hardware, leaving patients with non-functional medical devices they can no longer maintain (*IEEE Spectrum*)

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Thank you!

For spending the day building the Internet of Things with us.

Questions? Ask now, or write to me any time.



Slides, labs, and resources

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